Practice Quiz: Learning

Part A: Multiple Choice

1. The Dirichlet distribution is the conjugate prior for: A) Gaussian likelihood B) Categorical (multinomial) likelihood C) Poisson likelihood D) Exponential likelihood

2. When a Dirichlet prior $Dir(\alpha_1, \alpha_2, \alpha_3)$ is updated with counts (n_1, n_2, n_3), the posterior is: A) $Dir(\alpha_1 \cdot n_1, \alpha_2 \cdot n_2, \alpha_3 \cdot n_3)$ B) $Dir(\alpha_1 + n_1, \alpha_2 + n_2, \alpha_3 + n_3)$ C) $Dir(n_1, n_2, n_3)$ D) $N(\alpha_1 + n_1, \alpha_2 + n_2)$

3. The A matrix in a POMDP is learned by: A) Reinforcement learning with reward signals B) Accumulating Dirichlet concentration parameters that count state-observation co-occurrences C) Gradient descent on a loss function D) Random search

4. Bayesian Model Reduction (BMR) enables: A) Only parameter estimation B) Efficient structure learning by analytically comparing nested models without re-fitting C) Only forward model simulation D) Unsupervised clustering

5. The multivariate beta function $B(\alpha)$ appears in BMR because: A) It is the normalizing constant of the Dirichlet distribution B) It measures the mutual information between states C) It is the determinant of the A matrix D) It is the learning rate

6. As Dirichlet concentration parameters increase, the distribution: A) Becomes more diffuse (higher entropy) B) Becomes more concentrated (lower entropy) — reflecting greater confidence C) Stays unchanged D) Becomes bimodal

7. BMR is proposed to occur during: A) Active task performance B) Sleep — when the brain can evaluate model complexity without incoming data C) Eating D) Exercise

Part B: Short Answer

1. Starting from Dirichlet prior Dir(1, 1, 1) (uniform), compute the posterior after observing 100 data points with counts (60, 30, 10). What is the posterior mean? How does it compare to the MLE?
2. Explain why BMR is "analytic" — it does not require re-fitting the model to data. What mathematical property enables this efficiency?
3. In a POMDP with 5 states, an agent has learned transition probabilities with high confidence (large concentration parameters) for 4 states but very low confidence for the 5th state. Would BMR suggest pruning the 5th state? Explain the evidence calculation.